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CROP RECOMMENDATION SYSTEM

Using Machine Learning Techniques

AI-Based Smart Crop Prediction and Recommendation System

PRESENTATION 2026



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ABSTRACT OF THE PROJECT



Agriculture plays a vital role in the growth of our economy. Selecting the most suitable crop for cultivation is a crucial task for farmers to maximize yield and profit. However, crop selection depends on multiple factors such as soil nutrients, climate conditions, rainfall, temperature, humidity, and more.



This project, **"Crop Recommendation System Using Machine Learning Techniques"**, aims to develop a smart system that recommends the most suitable crop to be cultivated in a particular region based on soil and environmental parameters.



The system uses machine learning algorithms trained on historical datasets to predict the most appropriate crop. By providing accurate recommendations, the system helps farmers in better decision-making, optimizing resource usage, and increasing agricultural productivity.



The web-based application takes input parameters from users and predicts the best suitable crop along with confidence level, making agriculture more efficient and data-driven.



KEY BENEFITS

- ✓ Helps farmers choose the right crop
- ✓ Increases crop yield and profit
- ✓ Reduces risks of crop failure
- ✓ Promotes data-driven agriculture
- ✓ Supports sustainable farming practices



PROBLEM STATEMENT

Crop selection is a critical decision in agriculture that greatly influences productivity, profitability, and sustainability. However, farmers often face difficulties in choosing the right crop due to the following reasons:



Lack of Accurate Information

Farmers do not have access to accurate and complete information about soil nutrients, weather conditions, and crop suitability.



Unpredictable Environmental Factors

Variations in rainfall, temperature, humidity, and other climate conditions make crop selection uncertain.



Inadequate Soil Analysis

Soil nutrient levels (N, P, K, pH, etc.) vary from region to region, and manual analysis is time-consuming and error-prone.



Risk of Low Yield and Financial Loss

Selecting an unsuitable crop can lead to poor yield, increased costs, and significant financial losses for farmers.



Dependence on Traditional Methods

Many farmers still rely on traditional knowledge or experience, which may not always provide the best results.



IMPACT

- ➔ Reduced crop productivity and profitability
- ➔ Inefficient use of resources such as water, fertilizers, and land
- ➔ Food security challenges
- ➔ Hindrance to sustainable agriculture development



PROPOSED SOLUTION

To overcome the challenges in crop selection, we propose a “Crop Recommendation System Using Machine Learning Techniques”. This system analyzes soil and environmental parameters using machine learning algorithms and recommends the most suitable crop for a particular region.



1. Data Collection

Collect soil and environmental data such as N, P, K, pH, temperature, humidity, rainfall, and other relevant parameters.



2. Data Preprocessing

Clean and preprocess the collected data, handle missing values, and normalize the dataset for better model performance.



3. Model Training

Train machine learning models (e.g., Random Forest, Decision Tree, SVM, etc.) using historical crop data to learn patterns.



4. Crop Prediction

Use the trained model to predict the most suitable crop based on user input parameters.



5. Recommendation Output

Display the recommended crop along with confidence level and additional crop information to assist farmers.



KEY BENEFITS

- ✓ Provides accurate crop recommendations
- ✓ Improves crop yield and profitability
- ✓ Optimizes the use of resources (water, fertilizers, land)
- ✓ Reduces the risk of crop failure
- ✓ Supports data-driven and sustainable agriculture





SYSTEM ARCHITECTURE

The Crop Recommendation System follows a modular architecture that ensures efficient data flow, processing, model training, and accurate crop prediction.



1. Data Collection Layer

Collects soil and environmental data such as N, P, K, pH, temperature, humidity, rainfall, etc. from datasets and user input.



2. Data Preprocessing Layer

Handles missing values, removes noise, and normalizes the data to make it suitable for model training.



3. Model Training Layer

Machine learning algorithms are trained using the preprocessed dataset to learn patterns and relationships.



4. Prediction Layer

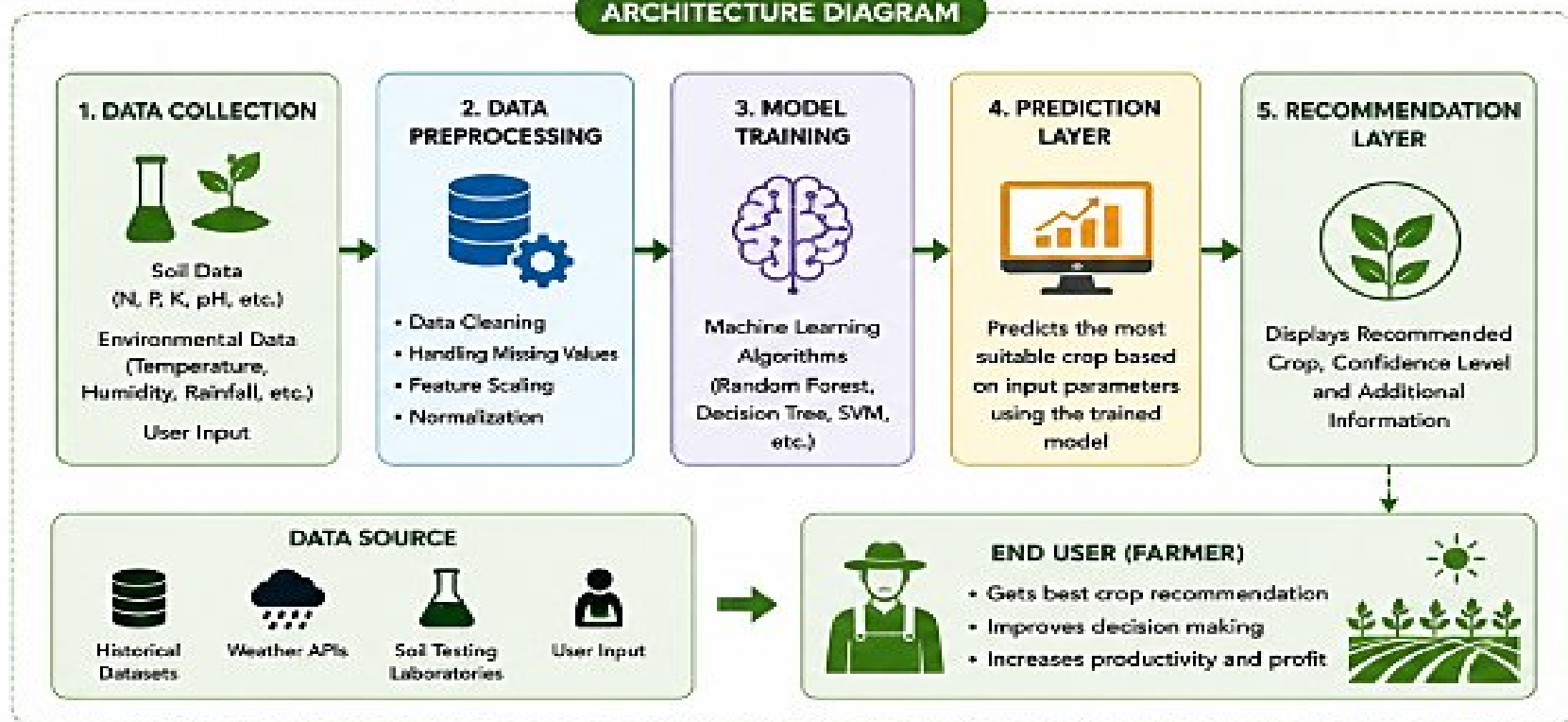
The trained model predicts the most suitable crop based on user input parameters.



5. Recommendation Layer

Displays the recommended crop along with confidence level and additional crop information to the user.

ARCHITECTURE DIAGRAM



TECHNOLOGIES USED



Python



scikit-learn



pandas

Flask
(Web Framework)HTML, CSS,
JavaScript

KEY FEATURES

- ✓ Accurate crop prediction using Machine Learning algorithms
- ✓ User-friendly web interface
- ✓ Supports multiple soil and environmental parameters
- ✓ Provides reliable and fast recommendations



MODEL TRAINING – PYTHON CODE



The following Python code demonstrates the training and evaluation of the Random Forest Classifier model for crop recommendation.

1 Import Libraries

- ▶ Import necessary modules for data splitting, Random Forest Classifier, and evaluation metric.

2 Features and Target

- ▶ Features (X) are all columns except 'label'.
- ▶ Target (y) is the 'label' column.

3 Split Dataset

- ▶ Split the dataset into training and testing sets (80% training, 20% testing) for model evaluation.

4 Train Model

- ▶ Initialize Random Forest Classifier with 100 trees and a fixed random state for reproducibility. Train the model using the training data.

5 Predict

- ▶ Use the trained model to predict the target values on the test set.

6 Accuracy

- ▶ Calculate and print the accuracy score of the model.

```
1  from sklearn.model_selection import train_test_split
2  from sklearn.ensemble import RandomForestClassifier
3  from sklearn.metrics import accuracy_score
4
5  # Features and Target
6  X = df.drop('label', axis=1)
7  y = df['label']
8
9  # Split Dataset
10 X_train, X_test, y_train, y_test = train_test_split(
11     X, y, test_size=0.2, random_state=42
12 )
13
14 # Train Model
15 model = RandomForestClassifier(n_estimators=100, random_state=42)
16 model.fit(X_train, y_train)
17
18 # Predict
19 y_pred = model.predict(X_test)
20
21 # Accuracy
22 accuracy = accuracy_score(y_test, y_pred)
23 print(f"Accuracy: {accuracy * 100:.2f}%")
```





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CONCLUSION

The Crop Recommendation System using Machine Learning Techniques successfully predicts the most suitable crop based on soil and environmental parameters. By leveraging data-driven insights, the system helps farmers make informed decisions, improve productivity, and ensure sustainable agriculture.



Key Takeaways



The system uses machine learning algorithms to analyze soil nutrients and environmental factors for accurate crop prediction.



It provides a user-friendly platform for farmers to get instant crop recommendations.



The solution reduces the risk of crop failure and enhances yield and profitability.



It promotes efficient use of resources such as water, fertilizers, and land.



The project contributes to smart farming and supports sustainable agricultural practices.



This project demonstrates the power of machine learning in agriculture and paves the way for a smarter, more sustainable future.



Overall Impact



Empowers farmers with data-driven crop recommendations.



Improves agricultural productivity and economic growth.



Supports sustainable farming and food security.



LIVE DEMO OF THE PROJECT

AI-Powered Crop Recommendation System Using Machine Learning

Project Demonstration

```
In [1]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [2]: import os

print(os.listdir('/content/drive/MyDrive'))
```

```
['1737253395301 (1).jpg', 'Copy of 1737253395301.jpg', 'IMG-20250127-WA0015.jpg', 'IMG-20250127-WA0013.jpg',
'MEFA IMP Q.docx', 'IMG_20250208_104529.jpg', 'my photo.jpg', 'DBMS UNIT - 3.docx', 'DBMS unit 1 (2).ppt',
'html_history.pdf', 'DBMS unit 2.ppt', 'DBMS unit 1 (1).ppt', 'DBMS unit 1.ppt', 'full stack development-1.docx',
'AAPAR ID.pdf', 'New Doc 06-12-2025 12.18_1.jpg', 'New Doc 06-12-2025 12.18_2.jpg', 'Pan card Deepika Tammireddy_1.jpg',
'Pan card Deepika Tammireddy_2.jpg', 'DOC-20250118-WA0011..zip', 'New Doc 08-18-2024 15.30.zip', 'Deepika.T identity card ISTS.zip',
'New Doc 06-08-2024 07.44(1).zip', 'New Doc 06-08-2024 07.53(1).zip', 'sridevi aadhar(1).zip', 'joint account.zip',
'Deepika.T identity card ISTS_2 (1).jpg', 'Deepika.T identity card ISTS_2.jpg', 'my aadhar.jpg', 'New Doc 06-12-2025 14.29_1.jpg',
'New Doc 06-12-2025 14.29_2.jpg', 'IMG_20250612_143110.jpg', 'IMG_20250612_145842.jpg', 'IMG_20250612_150009.jpg',
'2nd Yrs Python Full stack online(3).pdf', 'Day5 (1).pdf', 'DAY-15 (1).pdf', 'Day-16 (1).pdf', 'Infosys SB CourseEra Registered list_ISTS.xlsx',
'Day-14 (2).pdf', 'Day-13 (2).pdf', 'Day11&12 (2).pdf', 'day 10 (2).pdf', 'Day 9 (2).pdf', 'Day 8 (1).pdf', 'Day-7 15-05 (1).pdf',
'Day-7 (2).pdf', '2nd Yrs Python Full stack online.pdf', 'Day-6 (2).pdf', '2nd Python Full stack online.pdf', 'Day-14 (1).pdf',
'Day-13 (1).pdf', 'Day11&12 (1).pdf', 'day 10 (1).pdf', 'Day 9 (1).pdf', 'Day-7 15-05.pdf', 'Day-7 (1).pdf', 'Day-6 (1).pdf',
'12-CSE.pdf', 'Day-18 (1).pdf', 'Document from Ram Mohan (3)', 'Document from Ram Mohan (2)', 'Document from Ram Mohan (1)',
'Document from Ram Mohan', 'Day-16.pdf', 'DAY-15.pdf', 'Day11&12.pdf', 'DAY-19.pdf', 'Day-20.pdf', 'day 10.pdf', 'Day-7.pdf', 'Day 9.pdf',
'Day-17.pdf', 'Day5.pdf', 'day2&3.pdf', 'Day4.pdf', 'Day-6.pdf', 'Day 8.pdf', 'Day-18.pdf', 'Day-13.pdf', 'Day-14.pdf',
'Deepika Resume (2).pdf', 'Deepika Resume (1).pdf', 'Boolean.py', 'datatypesequence.py', 'Datatype']
```

GitHub Repository

Source Code & Project Documentation:

<https://github.com/TammireddyDeepika/AI-Powered-Crop-Recommendation-System-Using-Machine-Learning>



FUTURE SCOPE

The Crop Recommendation System has strong potential for future enhancements to make it more accurate, accessible, and beneficial for farmers and the agricultural ecosystem.



1. Improve Model Accuracy

Incorporate advanced machine learning and deep learning algorithms with larger datasets to improve prediction accuracy.



2. Real-time Weather Integration

Integrate real-time weather APIs and forecasts to provide dynamic crop recommendations.



3. Mobile Application Development

Develop a mobile application for easy access to crop recommendations, history tracking, and personalized farmer dashboard.



4. IoT and Sensor Integration

Use IoT sensors for real-time soil and environmental monitoring to provide more precise recommendations.



5. Multi-crop and Regional Expansion

Expand the system to recommend multiple crops for different seasons and include more regions and soil types.



These enhancements will make the system smarter, more reliable, and widely adopted, contributing to sustainable agriculture and food security.



6. Farmer Community and Expert Support

Build a farmer community platform and connect with experts for guidance and problem-solving.



7. Data Expansion and Continuous Learning

Continuously collect new data and implement model retraining for better performance over time.



8. Sustainability and Resource Optimization

Enhance recommendations by focusing on water conservation, soil health, and sustainable farming practices.





THANK YOU!



Thank you for your time and attention.

“

“Together, let's build a smarter and more sustainable future for agriculture.”

”



Crop Recommendation System
Using Machine Learning Techniques



Presented by:

Team Members

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Smart Decisions. Better Crops. Sustainable Future.

